Neha Manoj More

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Professional Summary

Data Scientist skilled in Python, SQL, statistical analysis and machine learning. Experienced in predictive modelling and automated analytics, with experience in building AI agents that simplify everyday tasks and make data easier to use.

Technical Skills

- **Machine Learning:** Regression, classification, clustering, demand prediction, feature engineering, scikit-learn, XGBoost, LightGBM and TensorFlow.
- **Statistics and Evaluation:** Probability and statistics, exploratory analysis, cross-validation, class imbalance, precision, recall, ROC-AUC, PR-AUC and threshold selection.
- **Geospatial Analysis:** Ward-level analysis, geographic identifiers, latitude and longitude, coordinate conversion, choropleth maps and Opendatasoft.
- **Programming and Data:** Python, pandas, NumPy, SQL, PySpark, PostgreSQL, SQL Server, Microsoft Fabric and Azure Data Factory.
- **ML Delivery and Collaboration:** Automated model workflows, saved models, NLP, LLM APIs, Git, Azure DevOps, CI/CD, testing and technical documentation.

Work Experience

Data Scientist **March 2025 – Present**
Leicester City Council *Leicester, UK*

- Develop predictive models to support service demand analysis and identify churn-risk groups using historical council data. Create ML features from service-use frequency, recency and previous activity to capture patterns relevant to prediction.
- Prepare ward-level geospatial analysis and choropleth maps to compare deprivation and population health across Leicester. Convert eastings and northings into latitude and longitude and prepare geographic fields for mapping and local-area analysis.
- Automate data preparation, model execution and output storage in Microsoft Fabric, connecting notebooks into repeatable workflows. Save selected models and evaluation outputs, with run identifiers and parameters to support comparison and investigation.
- Developed churn classification models and segmentation analysis to identify high risk users.
- Developed time series forecasting workflows using models like ARIMA, ARIMAX, SARIMA, and SARIMAX incorporating historical patterns and external variables to support demand forecasting and operational analysis.
- Implemented a Monte Carlo simulation for budget analysis, simulating future expenditure scenarios to estimate potential budget exhaustion timelines and quantify uncertainty around planning assumptions.
- Design and analyse experiments, including A/B testing, to compare outcomes and understand whether observed differences are statistically meaningful.
- Compare logistic regression, random forest and boosting models, using recall and PR-AUC to assess performance on imbalanced outcomes.
- Build an analysis workflow for 50,000+ citizen feedback records, covering text preparation, sentiment and topic classification. Use clustering to explore patterns in feedback and supervised models to assign categories for downstream reporting.
- Work with analysts and service colleagues to clarify questions, interpret results and make findings accessible through Power BI and clear written explanations.

Previous Experience

Associate Data Scientist **Aug 2020 – Jul 2023**
Tata Consultancy Services *India*

- Developed Python and SQL workflows to process healthcare and pharmaceutical data for analysis and business operations. Build and refine machine learning models to solve pharma data problems.
- Reduced the runtime of a SQL process from 10 minutes to 2 minutes, an 80% improvement, by optimising query logic.
- Combined records across PostgreSQL, Hadoop, Teradata and Salesforce to provide consistent inputs for downstream analysis.
- Built automated data integration workflows using Dell Boomi, IBM DataStage and SSIS.
- Applied validation and reconciliation checks to identify missing records and differences between source and processed data.
- Supported incremental loads and Kafka-based integrations to keep connected systems updated.
- Investigated processing failures and data issues, working with engineering teams to restore reliable outputs.
- Collaborated with business teams, developers and testers to clarify requirements, review changes and support releases using Git and CI/CD practices.

Projects

- **Urban Journey Duration Prediction** – *Planned portfolio project*
  Plan to predict taxi journey duration using public trip records, pickup and drop-off zones, departure time and calendar features. Compare a simple baseline with gradient boosting using a time-based test split.
- **Credit Risk Prediction** – *Python, scikit-learn, LightGBM*
  Prepared approximately 79,000 loan records and compared logistic regression, random forest and LightGBM models. Assessed precision, recall and ROC-AUC, including the effect of removing potentially problematic predictors.
- **Diabetic Retinopathy Detection** – *Python, TensorFlow, CNN*
  Developed an image classification model covering five disease severity grades. Used transfer learning and image augmentation, achieving 91% accuracy in project evaluation.
- **Document Question Answering** – *Python, LangChain, OpenAI API*
  Built a retrieval-augmented workflow using document chunking and vector embeddings to retrieve source material for LLM answers.

Education

MSc Advanced Computer Science **Sept 2023 – Dec 2024**
The University of Manchester, UK - Merit
Relevant study: Machine Learning, Data Science, Querying Data, Text Mining, Modelling Data on the Web, Data Visualization and Communication and Cyber Security.

BE Computer Science **Jul 2016 – Apr 2020**
Pune University, India
Relevant study: Data Givernance, Data Modelling, Statistics and Analytics, Machine Learning, Data Structures and Algorithms, Object-Oriented Programming.

Certifications

Microsoft Azure Data Scientist Associate · Microsoft Fabric Analytics Engineer · Microsoft Power BI Data Analyst · Microsoft AI Fundamentals · Microsoft Data Fundamentals